

UNIT 8: TREES, SPANNING TREES, AND GRAPH TRAVERSAL

Structure, Algorithms, Search Strategies, and Optimization

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Course Outline

- 1 Spanning Trees
- 2 Tree Traversal and Search Algorithms
- 3 Weighted Graphs
- 4 Minimum Spanning Tree Algorithms
- 5 Analysis
- 6 Summary



Outline

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Spanning Tree Definition

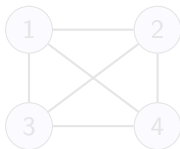
Definition

A spanning tree of a connected graph $G = (V, E)$ is a subgraph $T = (V, E')$ where:

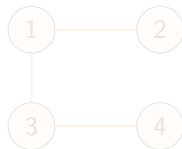
- T is a tree (connected and acyclic)
- T includes all vertices of G (spanning property)
- $|E'| = |V| - 1$ edges

less edges

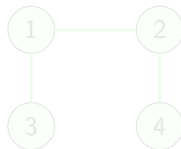
Original Graph (6 edges)



Spanning Tree 2 (3 edges)



Spanning Tree 1 (3 edges)

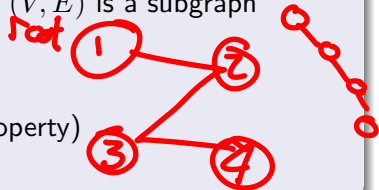


Spanning Tree Definition

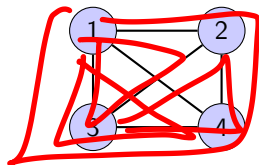
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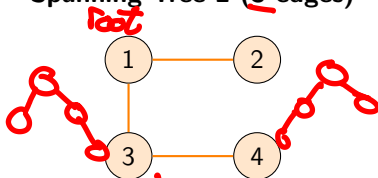
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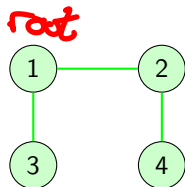
so much cycles



Spanning Tree 2 (3 edges)



~ distance



Spanning Tree 1 (3 edges)



Properties of Spanning Trees I

Structural Properties

- **Minimality:** Spanning tree with fewest edges that maintains connectivity
- **Uniqueness:** Not unique for most graphs (multiple spanning trees possible)
- **Edge count:** Always $|V| - 1$ edges
- **Connectivity:** Unique path between any two vertices

expansion

degree $\rightarrow$ 1-3

remove cycles



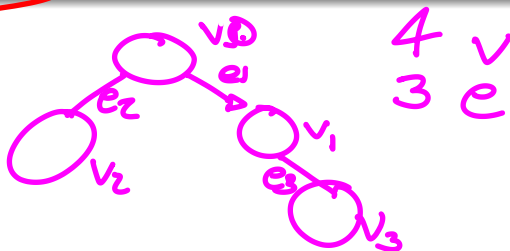
Properties of Spanning Trees II

Fundamental Property

Any set of $|V| - 1$ edges that forms a tree in G is a spanning tree if and only if all vertices are included.

Number of Spanning Trees

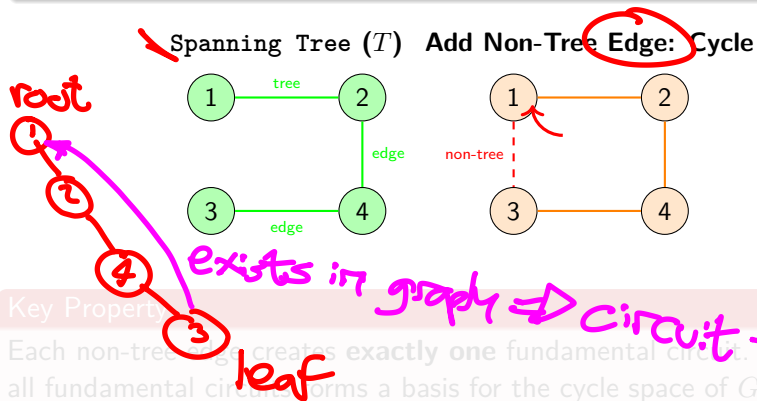
Computed using Kirchhoff's Matrix Tree Theorem. For K_n : n^{n-2} distinct spanning trees.



Fundamental Circuits

Fundamental Circuit

Adding a **non-tree edge** (edge not in spanning tree T) to T creates exactly one cycle, called a **fundamental circuit**.

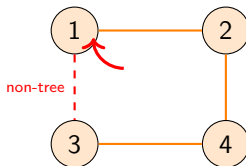
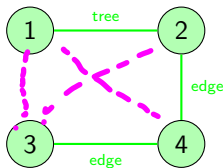


Fundamental Circuits

Fundamental Circuit

Adding a **non-tree edge** (edge not in spanning tree T) to T creates exactly one cycle, called a **fundamental circuit**.

Spanning Tree (T) Add Non-Tree Edge: Cycle



Key Property

Each **non-tree edge** creates **exactly one** fundamental circuit. The set of all fundamental circuits forms a basis for the **cycle space of G** .

→ "Kongo" → jerangura

- **Rank:** $r(G) = |V| - c$ where c is number of connected components
- **Nullity:** $\mu(G) = |E| - r(G) =$ number of independent cycles



Rank and Nullity

Rank and Nullity

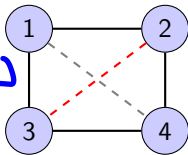
For a graph $G = (V, E)$:



- **Rank:** $r(G) = |V| - c$ where c is number of connected components
- **Nullity:** $\mu(G) = |E| - r(G) =$ number of independent cycles

$|V| = 4, |E| = 6, c = 1$

Original Graph

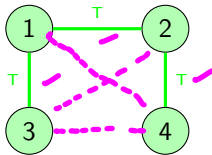


$$r(G) = |V| - c = 4 - 1 = 3$$

(tree edges)

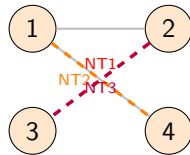
$|E'| = 3 = r(G)$

Spanning Tree



$\mu(G) = |E| - r(G) = 3$

Non-Tree Edges



$$\mu(G) = |E| - r(G) = 6 - 3 = 3$$

(non-tree edges)

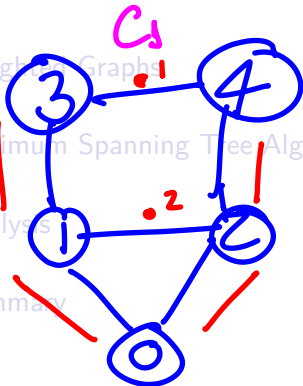
Outline

$$V = 7$$

$$E = 7$$

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出 入

degree

$$= (0) \rightarrow 2$$

$$(1) \rightarrow 3$$

$$(2) \rightarrow 3$$

$$(3) \rightarrow 2$$

$$(4) \rightarrow 2$$

$$(5) \rightarrow 1$$

$$(6) \rightarrow 1$$

can

$$C = 2$$

$$r(6) = 5$$

$$M(6) = 2$$

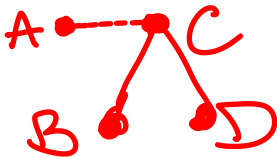
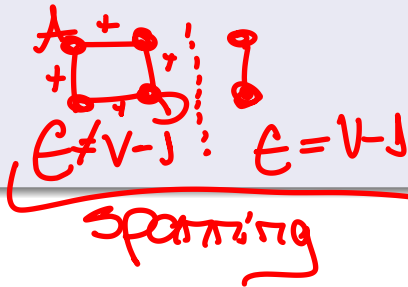


Graph Traversal: Overview

Motivation

Traversal algorithms systematically visit all vertices of a graph. Essential for:

- Finding connected components
- Detecting cycles
- Computing shortest paths
- Topological sorting



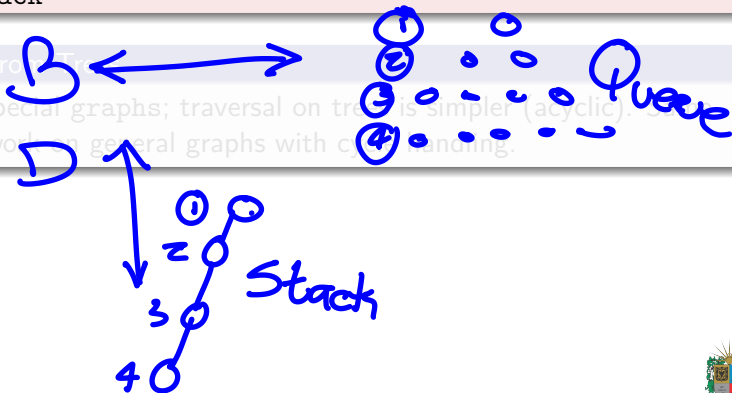
Graph Traversal: Overview

Two Primary Strategies

- Breadth-First Search (BFS): Explores layer-by-layer; uses queue
- Depth-First Search (DFS): Explores recursively down branches; uses stack

Distinction from Trees

Trees are special graphs; traversal on trees is simpler (acyclic). Search algorithms work on general graphs with cycle handling.



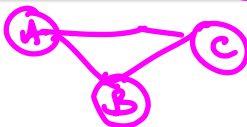
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Trees are special graphs; traversal on trees is simpler (acyclic). Same algorithms work on general graphs with cycle handling.



A → C
B → C

Pre
DFS



Breadth-First Search (BFS)

Algorithm Definition

BFS explores vertices in level order. Starting from a source vertex, visit all neighbors, then their neighbors, etc.

Data Structure: Queue (FIFO)

Pseudocode

distance
1
A | C B G | E D F

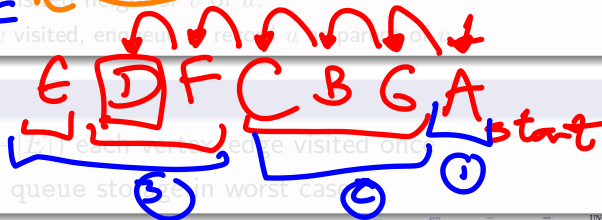
visited
↑
handle

source
target

Complexity

• **Time:** $O(|V| + |E|)$ each vertex/edge visited once

• **Space:** $O(|V|)$ queue storage in worst case



Breadth-First Search (BFS)

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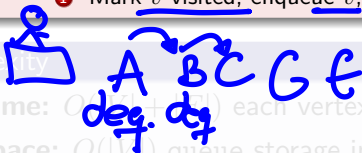
Data Structure: Queue (FIFO)

Pseudocode

- 1 Mark source as visited, enqueue it
- 2 While queue not empty:
 - 1 Dequeue vertex u
 - 2 For each unvisited neighbor v of u :
 - 1 Mark v visited, enqueue v , record u as parent of v

Complexity

- **Time:** $O(\sum_{v \in V} \deg(v))$ each vertex/edge visited once
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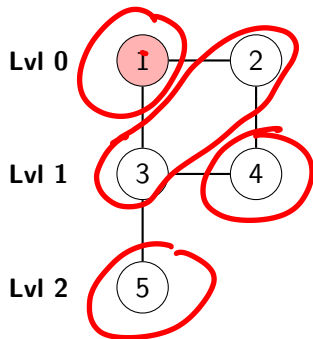
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Complexity

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BFS Visual Example



Queue (FIFO)

Start: [1]

Step 1: [2, 3]

Step 2: [3, 4]

Step 3: [4, 5]

Traversal Order

BFS from node 1: $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$

Visits all vertices at distance d before distance $d + 1$.



Depth-First Search (DFS)

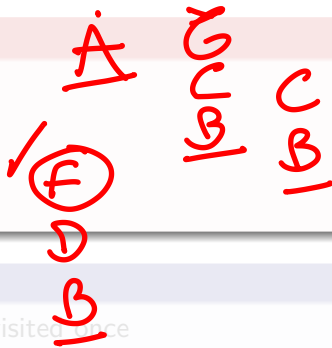
Algorithm Definition

DFS explores vertices by going deep into the graph. Recursively visit unvisited neighbors before backtracking.

Data Structure: Stack (LIFO) or recursion

Pseudocode (Recursive)

- 1 Mark u as visited
- 2 For each unvisited neighbor v of u :
 - 1 Record u as parent of v
 - 2 Recursively call DFS(v)



Complexity

- **Time:** $O(|V| + |E|)$ each vertex/edge visited once
- **Space:** $O(|V|)$ recursion stack in worst case

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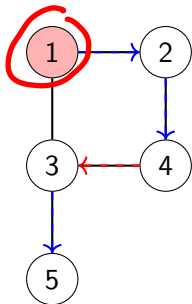
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Complexity

- **Time:** $O(|V| + |E|)$ each vertex/edge visited once
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DFS Visual Example



Stack (LIFO)

Start: [1]

Step 1: [2] ~~1~~ ³Step 2: [4] ~~2~~ ³Step 3: ~~3~~ 5

Traversal Order

DFS from node 1: 1 → 2 → 4 → 3 → 5

Goes as deep as possible before backtracking.

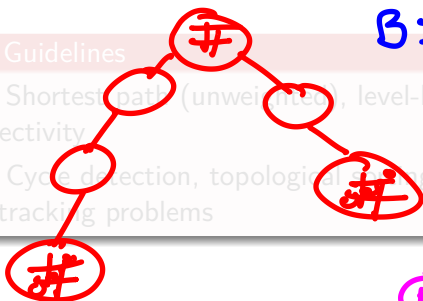
BFS vs. DFS Comparison

Comparison of Traversal Algorithms

Aspect	BFS	DFS
Time Complexity	$O(V + E)$	$O(V + E)$ ✓
Space Complexity	$O(V)$ (queue)	$O(V)$ (stack) ✓
Data Structure	Queue ✓	Stack/Recursion ✓
Visits by	Levels —	Depth first —
Shortest path	✓ (unweighted) •	✗ —
Memory efficient	Variable	Variable

Use Case Guidelines

- BFS: Shortest path (unweighted), level-based exploration, checking connectivity
- DFS: Cycle detection, topological sorting, connected components, backtracking problems



BFS



shortest 2



path = 3



BFS vs. DFS Comparison

Comparison of Traversal Algorithms

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Applications of Tree Traversal

BFS Applications

- Shortest path (unweighted)
- Level order traversal
- Connected components
- Bipartiteness check
- Social network *degrees*

DFS Applications

- Cycle detection
- Topological sort
- Strongly connected components
- Articulation points
- Backtracking: puzzles, mazes

Note

Both algorithms have identical time and space complexity but different traversal orders make them suitable for different problems.

hand

easy

machine

Tic-tac-toe
"video games
AI"



Applications of Tree Traversal

BFS Applications

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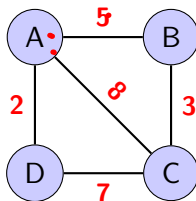
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Introduction to Weighted Graphs

Definition

A **weighted graph** assigns a numerical **weight** $w(e)$ to each edge e , representing cost, distance, capacity, or other attributes.



*Criterion
Optimization*

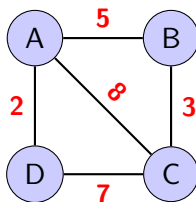
Weight Interpretation

- **Distance/Cost:** Edge weights represent real-world costs
- **Total Weight:** $w(G') = \sum_{e \in E'} w(e)$
- **Optimization Goal:** Find subgraph (path, tree, etc.) with minimum total weight

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F: FO

A B C

C

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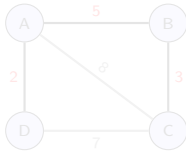
Engineering
↑
Opt.
↑
A

Minimum Spanning Tree (MST)

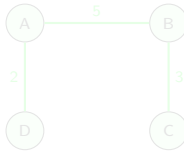
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A Minimum Spanning Tree of a weighted connected graph is a spanning tree with minimum total weight.

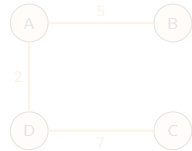
Graph (weight=25)



MST 1 (weight=10)



MST 2 (weight=12)



MST Properties

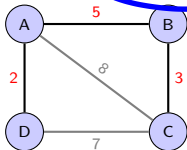
- Has minimum total weight among all spanning trees
- Always has $|V| - 1$ edges
- May not be unique if multiple edge weights are equal

Minimum Spanning Tree (MST)

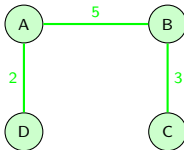
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A **Minimum Spanning Tree** of a weighted connected graph is a spanning tree with minimum total weight.

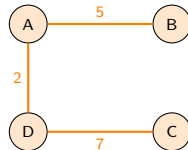
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MST Properties

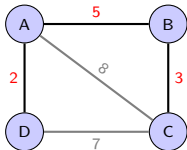
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Minimum Spanning Tree (MST)

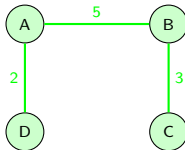
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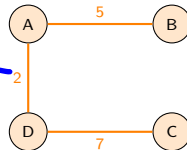
Graph (weight=25)



MST-1 (weight=10)



MST 2 (weight=12)



winner

MST Properties

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- Always has $|V| - 1$ edges
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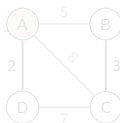


Prim's Algorithm

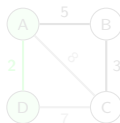
Algorithm Overview

Greedy approach: Start with a vertex and grow MST by adding minimum-weight edge connecting tree to non-tree vertices.

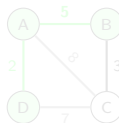
Step 1



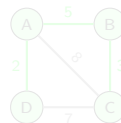
Step 2



Step 3



Complete



Complexity Analysis

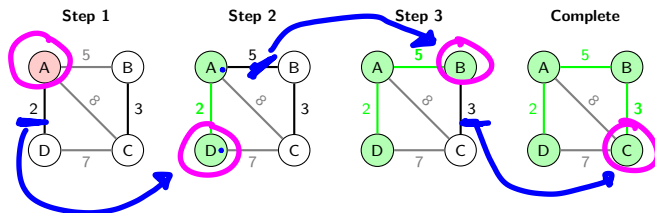
- Naive: $O(|V|^2)$ | With binary heap: $O((|V| + |E|) \log |V|)$



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Complexity Analysis

- Naive: $O(|V|^2)$ | With binary heap: $O((|V| + |E|) \log |V|)$



Kruskal's Algorithm

Algorithm Overview

Greedy approach: Sort edges by weight, add edges in order if they don't create a cycle (using Union-Find).

Sorted Edges: A-D(2), B-C(3), A-B(5), D-C(7)

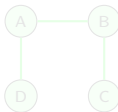
1. Add A-D



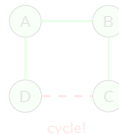
2. Add B-C



3. Add A-B



4. Skip D-C



Complexity Analysis

- **Total:** $O(|E| \log |E|)$ dominated by sorting + union-find operations



Kruskal's Algorithm

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A-C(8)

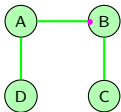
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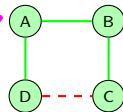
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cycle!

→ end

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Correctness of MST Algorithms

Cut Property

For any cut $(S, V \setminus S)$ of the graph, the minimum-weight edge crossing the cut must be in some MST.

Proof Strategy

- **Prim:** At each step, minimum-weight edge from tree to non-tree satisfies cut property
- **Kruskal:** Adding minimum-weight edge that doesn't create cycle respects cut property for its cut

Corollary

Both Prim's and Kruskal's algorithms produce optimal solutions. If all edge weights are distinct, MST is unique.



Correctness of MST Algorithms

Cut Property

For any cut $(S, V \setminus S)$ of the graph, the minimum-weight edge crossing the cut must be in some MST.

Proof Strategy

- **Prim:** At each step, minimum-weight edge from tree to non-tree satisfies cut property
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Algorithm Comparison

Prim vs. Kruskal Comparison

Aspect	Prim's	Kruskal's
Time Complexity	$O(V ^2)$ or $O((E + V) \log V)$	$O(E \log E)$
Space Complexity	$O(V)$ or $O(V + E)$	$O(V + E)$
Best for	Dense graphs	Sparse graphs
Data Structure	Priority queue	Sorting + Union-Find
Parallel	Difficult	More suitable

Selection Guideline

- $|E| \approx |V|^2$: Use Prim's (dense)
- $|E| \ll |V|^2$: Use Kruskal's (sparse)
- **Practical choice:** Kruskal's more commonly implemented due to simplicity



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Complexity Considerations

Practical Factors

- **Graph density:** Determines dominance of edge vs. vertex operations
- **Data structures:** Choice of implementation (array, heap, union-find) affects constants
- **Edge weight distribution:** Affects number of comparisons in practice
- **Memory access patterns:** Cache locality impacts real-world performance



Complexity Considerations

Advanced Techniques

- **Borůvka's algorithm:** Parallel-friendly MST with $O(|E| \log |V|)$ complexity
- **Randomized approaches:** Expected linear time under certain conditions
- **Approximation methods:** For very large graphs requiring approximate MSTs



Outline

- 1 Spanning Trees
- 2 Tree Traversal and Search Algorithms
- 3 Weighted Graphs
- 4 Minimum Spanning Tree Algorithms
- 5 Analysis
- 6 Summary



Program Topics Covered

✓ Spanning Trees

Definition and properties

✓ Rank and Nullity

Graph cycle structure

✓ Fundamental Circuits

Cycle generation from spanning trees

✓ Tree Traversal

BFS and DFS algorithms

✓ Weighted Graphs

Edge weights and costs

✓ MST Algorithms

Prim's and Kruskal's



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-ii>

